

Altera Cyclone IV EP4CE6 Development Board (V3.0)

Hardware Architecture & Peripheral Pin Reference

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1 System Architecture Overview

The Cyclone IV V3.0 board is built around an Altera EP4CE6 FPGA in a 144-pin Quad Flat Package (QFP). To support standalone operation without requiring external peripheral shields, the board integrates essential human-machine interface (HMI) components, secondary non-volatile flash storage, and regulated low-dropout (LDO) power supplies.

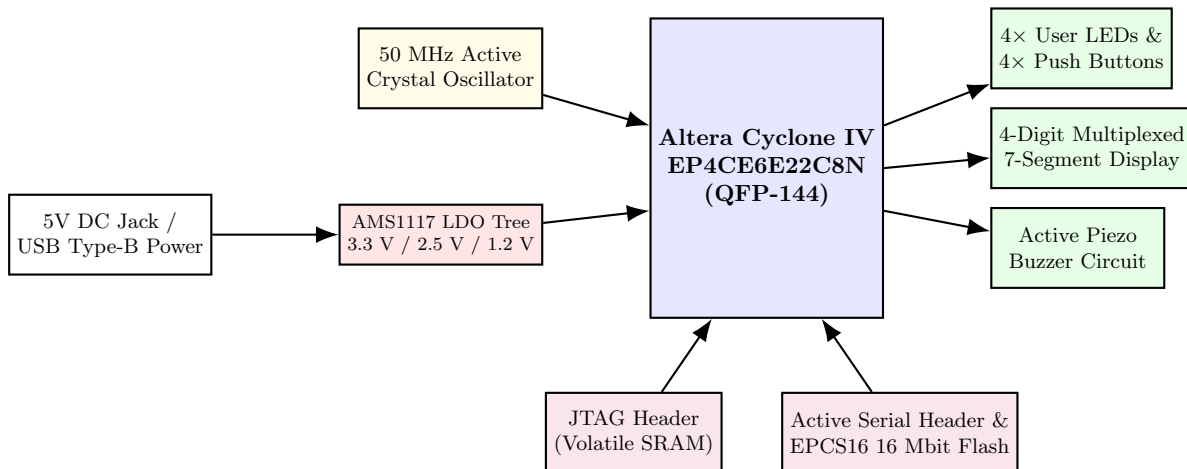


Figure 1: System-level hardware block diagram of the Altera Cyclone IV V3.0 development board.

2 Core FPGA Specifications (EP4CE6E22C8N)

The target FPGA belongs to the low-power Cyclone IV E series. The device nomenclature decodes as follows: **EP4C** (Cyclone IV architecture), **E** (Enhanced logic/low power), **6** ($\approx 6,000$ logic elements), **E22** (EQFP-144 package), **C8** (Commercial temperature grade 0°C to 85°C , speed grade 8), and **N** (Lead-free RoHS compliant).

3 Configuration & Programming Interfaces

The V3.0 board features two distinct 10-pin programming headers configured for use with Altera USB-Blaster download cables:

1. **JTAG (Joint Test Action Group) Interface**
2. **Active Serial (AS) Interface:** Writes configuration data directly into the onboard **EPCS16** (16 Mbit / 2 MByte) serial flash memory chip.

4 Onboard Peripherals & Pin Assignments

Peripheral Component	Board Label	FPGA Pin Assignment	Logic Standard / Notes
System Clock (50 MHz)	CLK50	PIN_23	3.3-V LVTTL (Global Clock Input)
User LEDs (Active Low)	LED1	PIN_87	3.3-V LVTTL (0 = Illuminated)
	LED2	PIN_86	
	LED3	PIN_85	
	LED4	PIN_84	
Push Buttons (Active Low)	KEY1 / S1	PIN_88	3.3-V LVTTL (0 = Pressed, Debounce req.)
	KEY2 / S2	PIN_89	
	KEY3 / S3	PIN_90	
	KEY4 / S4	PIN_91	
Piezo Buzzer	BUZZER	PIN_110	3.3-V LVTTL (1 = Tone Active)
7-Segment Anodes (Digit Sel.)	DIG1 (MSB)	PIN_133	3.3-V LVTTL (0 = Digit Enabled)
	DIG2	PIN_132	
	DIG3	PIN_131	
	DIG4 (LSB)	PIN_128	
7-Segment Cathodes (Data)	SEG_A	PIN_127	3.3-V LVTTL (0 = Segment Lit)
	SEG_B	PIN_126	
	SEG_C	PIN_125	
	SEG_D	PIN_124	
	SEG_E	PIN_121	
	SEG_F	PIN_120	
	SEG_G	PIN_119	
	SEG_DP	PIN_115	

Table 1: Quartus Prime physical pin mapping reference for Altera Cyclone IV V3.0 onboard peripherals.

5 7-Segment Display Lookup Table

Segment Bitmap Mapping

The 8-bit output vector maps to the 7-segment display segments as follows (active low: 0 = ON):

Bit Index	Physical Segment
Bit 0 (LSB)	Top segment
Bit 1	Top-left segment
Bit 2	Bottom-left segment
Bit 3	Bottom segment
Bit 4	Bottom-right segment
Bit 5	Top-right segment
Bit 6	Middle segment
Bit 7 (MSB)	Decimal point (DP)

Hardware Decoder Truth Table

This lookup table compares the logical binary code (readable: 1 = ON) with the actual inverted physical output used in the testbench (0 = ON).

Char	4-bit Input	Logical Binary (1 = ON)	Physical Output (0 = ON)
0 (ZERO)	0000	00111111	11000000
1 (ONE)	0001	00000110	11111001
2 (TWO)	0010	01011011	10100100
3 (THREE)	0011	01001111	10110000
4 (FOUR)	0100	01100110	10011001
5 (FIVE)	0101	01101101	10010010
6 (SIX)	0110	01111101	10000010
7 (SEVEN)	0111	00000111	11111000
8 (EIGHT)	1000	01111111	10000000
9 (NINE)	1001	01101111	10010000
A	1010	01110111	10001000
b	1011	01111100	10000011
C	1100	00111001	11000110
d	1101	01011110	10100001
E	1110	01111001	10000110
F	1111	01110001	10001110

Table 2: Lookup table mapping 4-bit hexadecimal inputs to inverted 8-bit segment outputs.